



Introduction

Edge AI applications like video analytics and speech recognition require massive numbers of MAC operations, but most accelerators use fixed precision and support only CNN or RNN workloads, wasting power and limiting flexibility. This work proposes a low-power, reconfigurable MAC accelerator that supports both CNN and RNN processing with runtime precision scaling and selective core activation, enabling energy-efficient AI computation on edge devices.

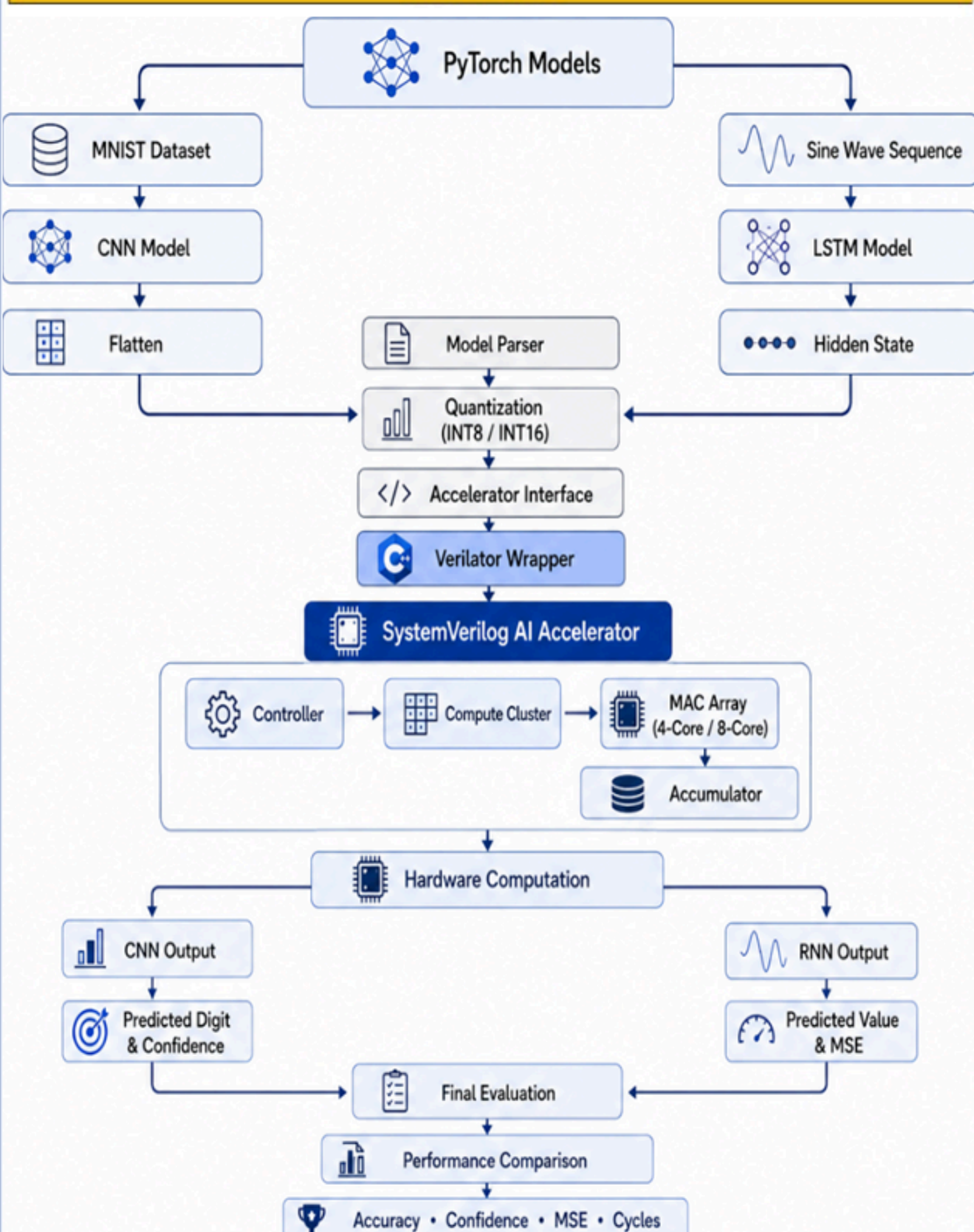
Problem Definition

Current neural accelerators lack a unified, runtime-adaptive architecture that seamlessly adjusts arithmetic precision and computational mode for both CNN and RNN workloads, resulting in sub-optimal power-performance and reduced efficiency.

Objectives

- Design a unified MAC accelerator supporting both CNN (spatial) and RNN (sequential) processing
- Enable precision adaptability to switch between INT16 and INT8 within the same hardware
- Support scalable parallelism by activating only the required number of MAC cores
- Evaluate how these adaptations impact overall power consumption

Methodology

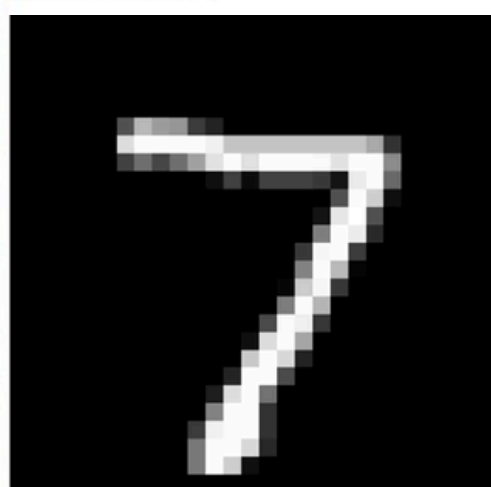


Tools used



Results and Discussions

The proposed accelerator efficiently runs both CNN and RNN workloads on a single adaptive datapath with dynamic precision and core scaling. Results show minimal accuracy loss in INT8 mode and noticeable power savings with reduced precision and active cores.



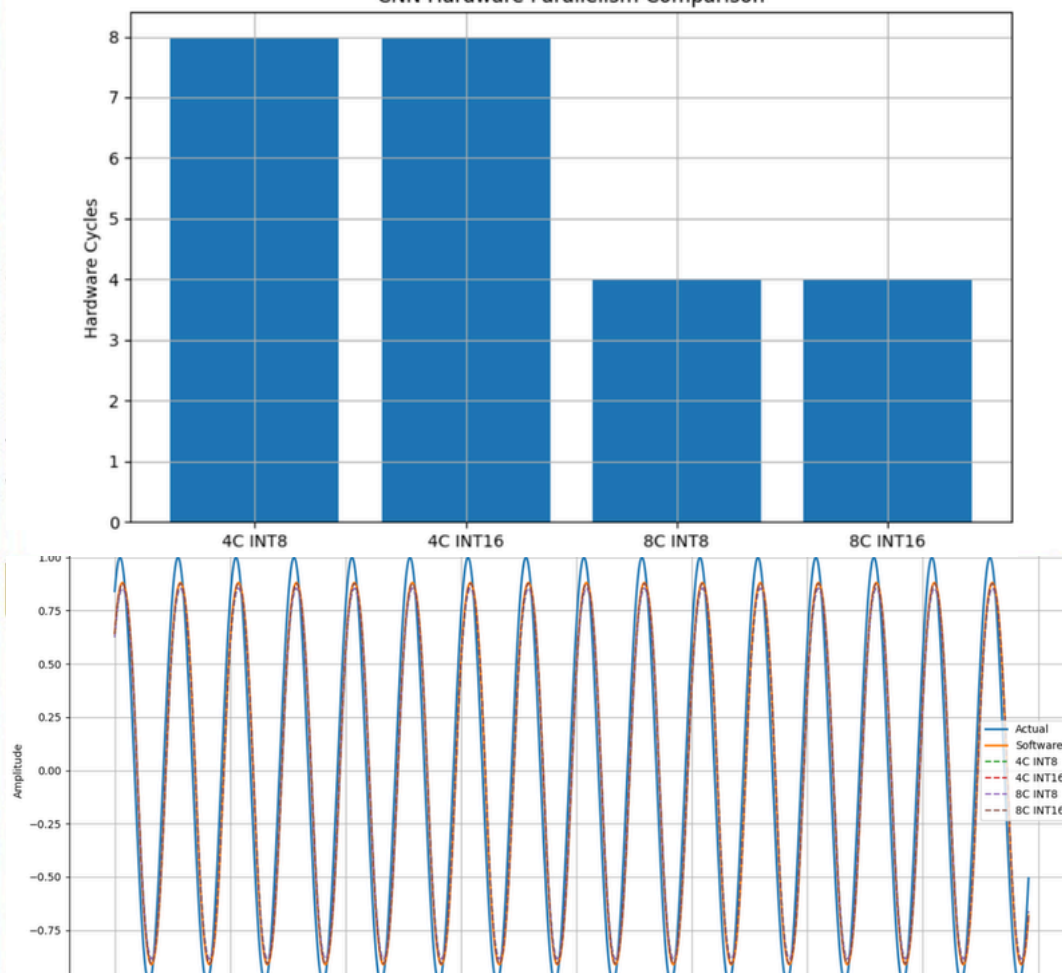
Hardware Assisted

4C INT8
Prediction : 7
Confidence : 99.99984503%
Cycles : 100352

4C INT16
Prediction : 7
Confidence : 99.99983311%
Cycles : 100352

8C INT8
Prediction : 7
Confidence : 99.99984503%
Cycles : 50176

CNN Hardware Parallelism Comparison



Conclusions

This work demonstrates a low-power, reconfigurable MAC-based accelerator that unifies CNN and RNN processing within a single adaptive datapath. By combining precision scaling and selective core activation, the design achieves energy-efficient computation with minimal accuracy degradation, making it well suited for flexible and power-constrained edge AI applications.